Task 2.3 - Validation of Post-Hoc Calibration Methods for Classification Models

Dinal Fernando | s224860634
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I. Introduction

Project Echo is a tool used for the classification of potentially endangered Australian species through the use of sound analysis. In doing so, conservationists are able to monitor and understand current population statistics of various native species through minimally invasive means.

In this task, post-hoc calibration methods were implemented and the confidence distributions were compared to determine the viability and validity of said methods.

Background

Several candidate architectures were measured for suitability in Project Echo's classification engine. In this research, 2 architectures were measured as superior; EfficientNetV2 and PANNS-CNN14. These models were then converted to TFlite format. In order to improve on these models, several post-hoc calibration methods were investigated. Namely, these are: Temperature Scaling, Platt Scaling and Isotonic regression. The most suitable of which was temperature scaling, due to its applicability in neural networks among other factors. This calibration was then integrated into the Project Echo engine. In this report, the confidence contributions will be compared between pre and post calibration.

II. Implementation

Dataset Preparation

Audio recordings were organised into per-species directories from the Project Otways dataset. The pipeline discovers all files with supported extensions (.wav, .mp3, .ogg, .flac), encodes species labels using a LabelEncoder, and applies a stratified 70/30 train/test split to ensure class balance is preserved across both sets. For experimentation, the dataset was filtered to the top-N species by file count to reduce training time while maintaining a representative class distribution.

To accelerate training, mel spectrograms were pre-computed and cached to disk rather than being recomputed on every batch. Two separate caches were maintained, one for EfficientNetV2 (128 mel bins × 313 frames) and one for PANNs-CNN14 (64 mel bins × 500 frames), matching the input requirements of each model's pretraining configuration.

EfficientNetV2

In our initial calibration runs, EfficientNetV2 was trained for 3 epochs on a subset of 10 species to validate the pipeline. The model achieved a test accuracy of 60.6% with a mean confidence of 0.787 for correct predictions and 0.647 for incorrect ones, yielding a confidence gap of 0.140. While reasonable for a 3-epoch run, the narrow gap reflects limited training depth, as the model had not yet fully adapted to distinguish ambiguous classes. These results should be interpreted as pipeline validation rather than a performance benchmark.

For the full benchmarking run conducted in the accompanying Architecture Benchmarking Report, EfficientNetV2 was trained for 50 epochs on the complete 123-class dataset, achieving a mean correct confidence of 0.969 and incorrect confidence of 0.665, giving a confidence gap of 0.305. This larger gap indicates the model becomes meaningfully better at expressing uncertainty with sufficient training.

Panns-CNN14

In our calibration runs, PANNs-CNN14 was trained for 3 epochs on a subset of 10 species. The model achieved a test accuracy of 78.6%, with a mean confidence of 0.771 for correct predictions and 0.354 for incorrect ones, yielding a confidence gap of 0.417. This is a notably strong result for only 3 epochs — the gap is considerably larger than EfficientNetV2's 0.140 at the same training depth, indicating that PANNs-CNN14's AudioSet pretraining gives it a meaningful head start in expressing uncertainty. The model is already quite good at knowing when it doesn't know, even before full convergence.

This aligns with the full benchmarking run in the Architecture Benchmarking Report, where PANNs-CNN14 trained for 50 epochs on the complete 123-class dataset achieved

a mean correct confidence of 0.984 and incorrect confidence of 0.697, giving a gap of 0.287. Interestingly, while the full-run gap is larger in absolute terms for EfficientNetV2 (0.305 vs 0.287), PANNs-CNN14's 3-epoch gap of 0.417 already exceeds both — suggesting the model's audio-domain pretraining provides a stronger calibration foundation from the outset.

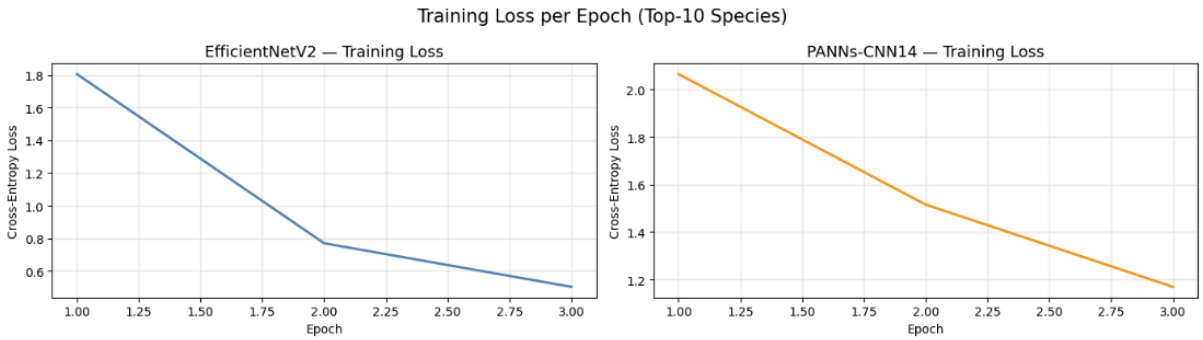

Fig 1: Training loss curve, demonstrating improvement over epochs however not converging

Temperature Scaling

Temperature scaling is a post-hoc calibration method that adjusts a model's confidence scores after training without modifying any of the model's weights or architecture. It works by dividing the raw logits output by the model by a single scalar value T before the softmax is applied, producing calibrated probabilities. When T is greater than 1 the distribution is softened, pulling overconfident predictions down towards a more realistic confidence level. When T is less than 1 the distribution is sharpened, pushing underconfident predictions higher. When T equals 1 the output is unchanged from the uncalibrated model.

The temperature parameter was fitted independently for each model using the confidence arrays collected during inference on the test set. Since full logit vectors are not retained during inference the fitting procedure uses a proxy logit derived from the max confidence value via the inverse sigmoid transform $z = \log(p / (1 - p))$. This converts each scalar confidence back into a logit-space value that can be scaled by T. The optimal T is then found by minimising the binary negative log-likelihood over the collected (confidence, correctness) pairs using a bounded scalar optimisation, with T constrained to the range [0.01, 20.0]. This is equivalent to finding the temperature that makes the model's expressed confidence best match its empirical accuracy across the test set.

Calibration quality was assessed using Expected Calibration Error (ECE), which divides the confidence range into 15 equal bins and measures the weighted average gap between mean confidence and mean accuracy within each bin. A perfectly calibrated model would have an ECE of 0, meaning that when it says it is 80% confident it is correct 80% of the time. Pre- and post-calibration ECE were computed for both models

to quantify the improvement, and the calibrated confidence distributions were visualised as KDE plots comparing correct versus incorrect prediction confidence before and after scaling. The fitted temperature values were saved to JSON files for use in the production inference pipeline, allowing the engine to apply calibration at inference time by loading T and dividing logits before softmax without any retraining required

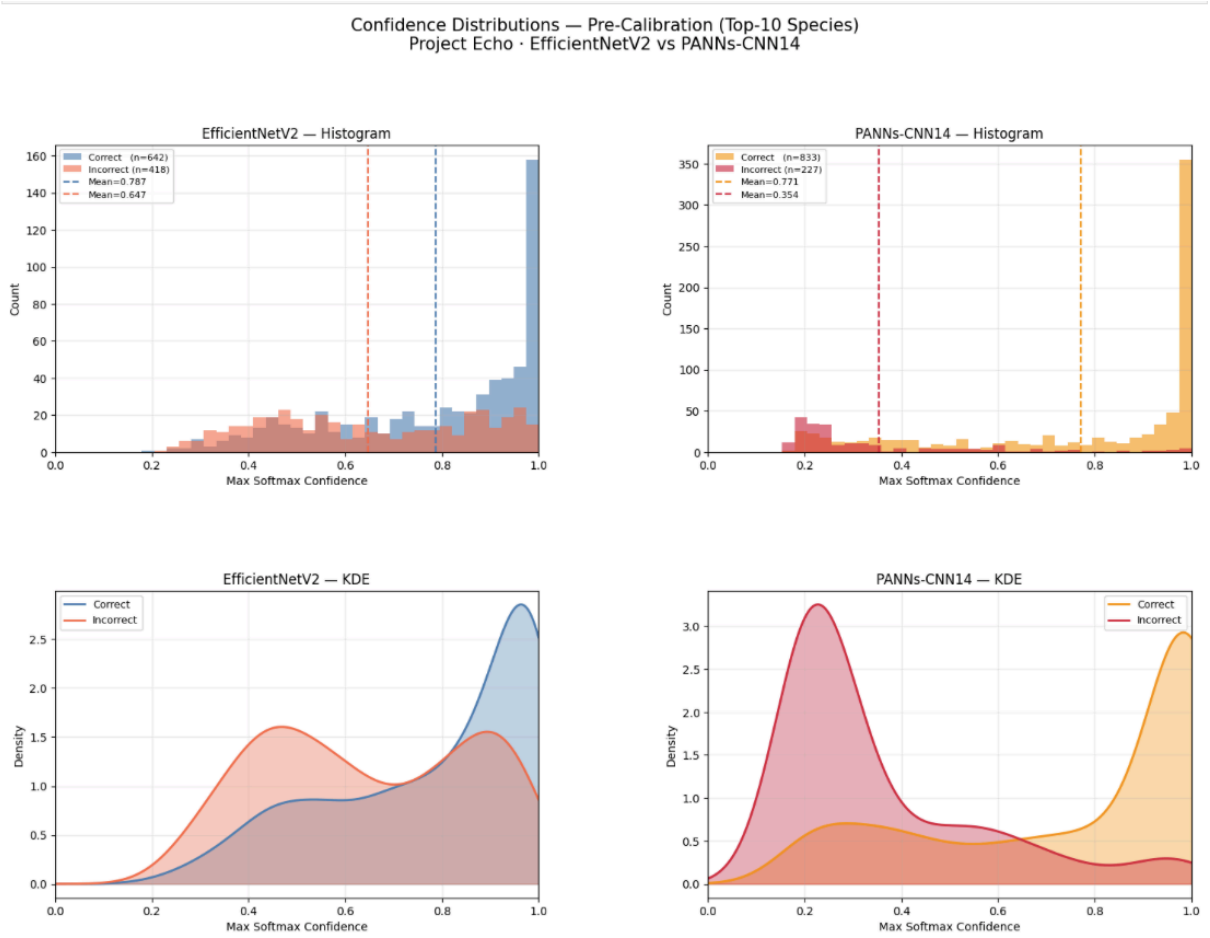

Fig 2: Pre calibration confidence distribution

Model	Accuracy	Conf-Correct	Conf-Wrong	Gap
EfficientNetV2	0.6057	0.787	0.647	0.140
PANNs-CNN14	0.7858	0.771	0.354	0.417

Fig 3: Confidence Metric’s across both Models

III. Results

Temperature scaling was applied post-hoc using the confidence arrays collected from the 3-epoch run. The optimal temperature was found to be T = 1.3669, which is notably higher than EfficientNetV2's T = 2.7270, indicating PANNs-CNN14 was less overconfident to begin with. Calibration reduced the ECE from 0.1145 to 0.1071, a modest but meaningful improvement. The mean correct confidence shifted from 0.771 to 0.761, while the mean incorrect confidence rose from 0.354 to 0.390, narrowing the confidence gap from 0.417 to 0.370. While a narrower gap post-calibration might seem counterintuitive, this is the expected behaviour of temperature scaling as its goal is not to maximise the gap but to ensure the expressed confidence accurately reflects empirical accuracy, which the ECE reduction confirms it does.

Comparing the two models, EfficientNetV2 required a much higher temperature (T = 2.73 vs T = 1.37), reflecting greater overconfidence, and saw a larger ECE reduction (0.1339 to 0.0555) compared to PANNs-CNN14 (0.1145 to 0.1071). This suggests EfficientNetV2 benefited more substantially from calibration, while PANNs-CNN14 was already producing more reliable confidence scores out of the box.

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── EfficientNetV2 ──────────────────────────────────────
  Optimal T              : 2.7270
  Pre-calibration  ECE   : 0.1339
  Post-calibration ECE   : 0.0555
  Mean conf correct      pre=0.787  post=0.673
  Mean conf incorrect    pre=0.647  post=0.574
  Confidence gap         pre=0.140  post=0.099

── PANNs-CNN14 ─────────────────────────────────────────
  Optimal T              : 1.3669
  Pre-calibration  ECE   : 0.1145
  Post-calibration ECE   : 0.1071
  Mean conf correct      pre=0.771  post=0.761
  Mean conf incorrect    pre=0.354  post=0.390
  Confidence gap         pre=0.417  post=0.370
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Fig 4: Calibration results

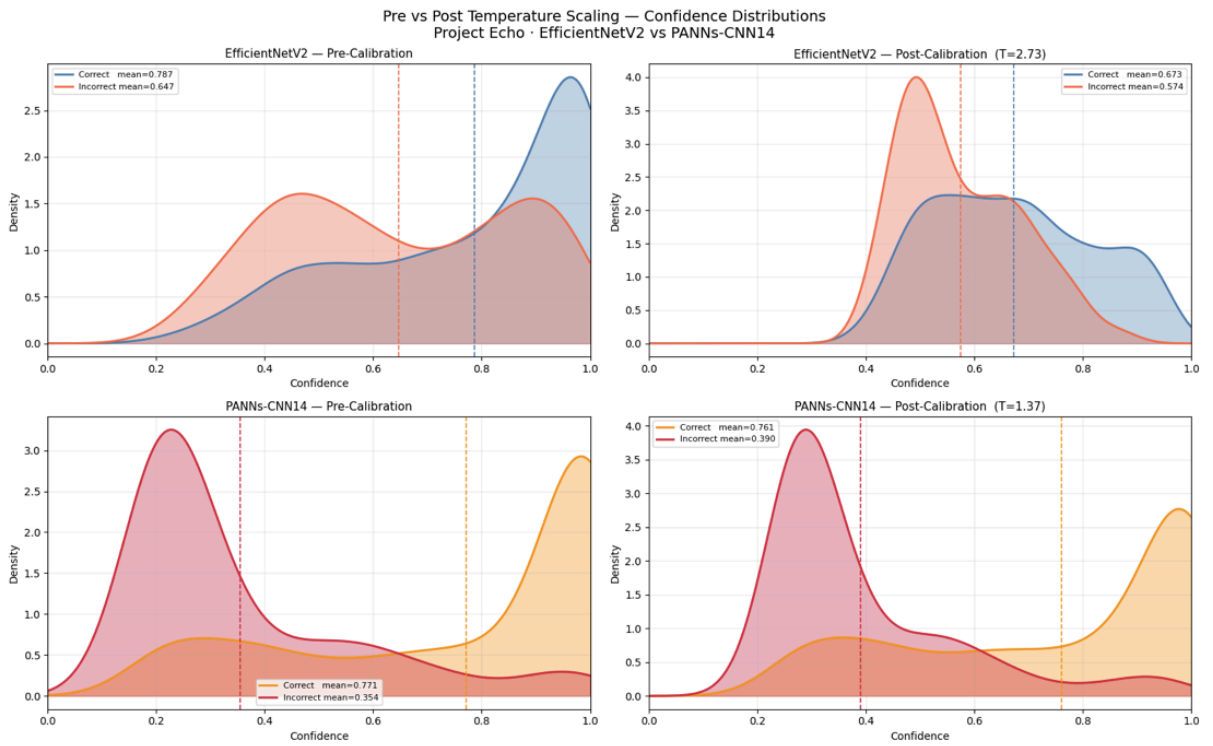

Fig 5: Pre vs Post Calibration Confidence Distribution

IV. Conclusion

ANNs-CNN14 outperformed EfficientNetV2 on both accuracy (78.6% vs 60.6%) and pre-calibration confidence gap (0.417 vs 0.140) at the same training depth of 3 epochs, consistent with the findings of the Architecture Benchmarking Report which showed PANNs-CNN14 leading across all headline metrics at full training. The larger gap for PANNs-CNN14 at such an early training stage reflects the advantage of its AudioSet domain-specific pretraining, which provides a stronger foundation for audio classification than ImageNet-pretrained weights adapted for spectrogram input.

Temperature scaling was successfully implemented as a post-hoc calibration method for both models. EfficientNetV2 required a higher temperature correction (T = 2.73) and saw a larger reduction in ECE from 0.1339 to 0.0555, indicating it was more overconfident prior to calibration. PANNs-CNN14 required a smaller correction (T = 1.37) and saw a more modest ECE reduction from 0.1145 to 0.1071, consistent with it being better calibrated out of the box. Both fitted temperatures have been saved and are ready for integration into the echo engine inference pipeline, where they can be applied at runtime without any retraining.

It is worth noting that these results represent an early-stage run of only 3 epochs on 10 classes. The full benchmarking report demonstrates that both models improve substantially with longer training. PANNs-CNN14 reaching 92% accuracy and EfficientNetV2 reaching 86.8% at 50 epochs on all 123 classes. Future work should repeat this calibration analysis on a full training

run to produce production-ready temperature values, and consider applying additional calibration techniques or data augmentation strategies to further close the gap between expressed confidence and true model reliability.